



CHEMISTRY

8872/01

Paper 1 Multiple Choice

Wednesday

19th SEPTEMBER 2012

50 minutes

Additional materials: Data Booklet
 Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write in soft pencil.

Do not use staples, paper clips, highlighters, glue or correction fluid.

Write your name, index number and CG on the Answer Sheet in the spaces provided.

There are **thirty** questions on this paper. Answer **all** questions. For each question, there are four possible answers labelled **A, B, C, and D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Section A

For each question, there are four possible answers, **A, B, C and D**. Choose the **one** you consider to be correct.

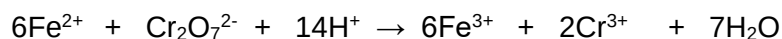
- 1 If a mixture of 0.4 g hydrogen and 9.6 g oxygen is exploded, what will be the volume of the remaining gas at room temperature and pressure?

A 0.60 dm³ **B** 1.20 dm³ **C** 2.40 dm³ **D** 4.80 dm³

- 2 Which of the following contains two isoelectronic species?

A NH₄⁺ and BH₃
B NH₄⁺ and CH₄
C BF₃ and NH₃
D PF₅ and BF₄⁻

- 3 Iron(II) ions, Fe²⁺, are oxidised by aqueous acidified potassium dichromate(VI) according to the equation:



50.0 cm³ of an iron(II) solution require 45.20 cm³ 0.100 mol dm⁻³ acidified potassium dichromate(VI) for reaction, what is the mass of Fe²⁺ ion in 1 dm³ of the iron(II) solution?

A 12.6 g
B 25.2 g
C 30.3 g
D 60.5 g

- 4 Why is BCl₃ molecule planar, whereas PH₃ molecule is pyramidal?

A The lone pair – bond pair electron repulsion is greater in PH₃ than that of BCl₃.
B The boron atom in BCl₃ has six electrons in its outermost shell, whereas the phosphorus atom in PH₃ has eight electrons in its outermost shell.
C The repulsion between chlorine atoms is greater than that between hydrogen atoms.
D The electronegativity of chlorine is greater than that of hydrogen.

- 5 All of the carbon dioxide in a soft drink with an initial mass of 81.04 g was carefully extracted and collected as a gas. The final mass of the drink was 80.41 g.

What volume would the carbon dioxide occupy at room conditions?

- A 0.14 dm³ B 0.32 dm³
C 0.34 dm³ D 0.63 dm³

- 6 The successive ionisation energies (IE) of two elements **X** and **Y**, are given below:

IE/ kJ mol ⁻¹	1 st	2 nd	3 rd	4 th	5 th	6 th	7 th	8 th
X	786	1580	3230	4360	16000	20000	23600	29100
Y	1251	2298	3822	5158	6542	9330	11000	33604

The compound formed when **X** and **Y** combine is most likely to be

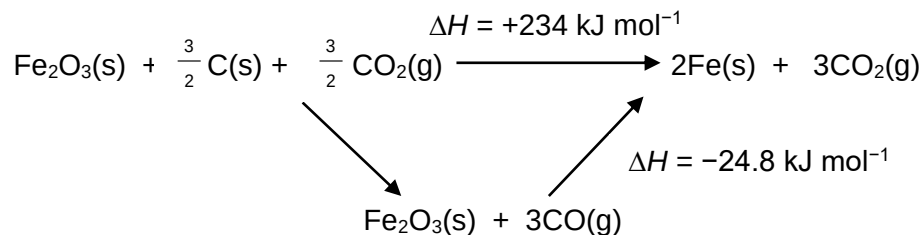
- A ionic, with formula **X₂Y**
B ionic, with formula **XY₂**
C covalent, with formula **XY₄**
D covalent, with formula **X₂Y₅**
- 7 Beryllium resembles aluminium in its chemical properties.

Which property of beryllium compounds is **not** correct?

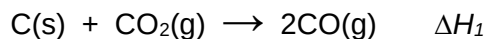
- A Beryllium chloride is an ionic compound.
B Beryllium chloride can form the dimer Be₂Cl₄.
C Beryllium oxide can undergo reaction with aqueous potassium hydroxide.
D Beryllium chloride dissolves in water to give an acidic solution.
- 8 Which of the following correctly describes the types of structures present in the Period 3 oxides listed below?

	Na ₂ O	SiO ₂	P ₄ O ₁₀	SO ₃
A	giant ionic	giant molecular	giant molecular	simple molecular
B	giant ionic	giant molecular	simple molecular	simple molecular
C	giant ionic	giant ionic	giant molecular	simple molecular
D	giant ionic	giant ionic	simple molecular	simple molecular

- 9 The following energy cycle is established.



Using the energy cycle, calculate ΔH_1 for the following reaction:

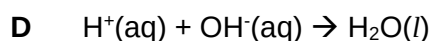
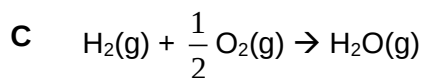
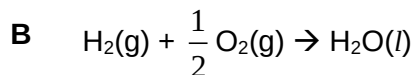
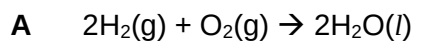


- A +86.3 kJ mol⁻¹
 B +139.5 kJ mol⁻¹
 C +172.5 kJ mol⁻¹
 D +258.8 kJ mol⁻¹
- 10 When 10 cm³ of 1 mol dm⁻³ H₂SO₄(aq) was added to a 50 cm³ of 1 mol dm⁻³ KOH(aq), a temperature rise of 5 °C was obtained.

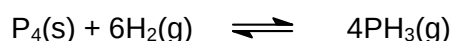
Assuming that there is negligible heat loss to the surroundings, what is the enthalpy change of neutralisation? (Density of solution may be taken as 1 g cm⁻³ and specific heat capacity of the solution as 4.2 J g⁻¹ K⁻¹)

- A $-\frac{60 \times 4.2 \times 5 \times 1000}{20} \text{ kJ mol}^{-1}$
 B $-\frac{60 \times 4.2 \times 5 \times 1000}{10} \text{ kJ mol}^{-1}$
 C $-\frac{60 \times 4.2 \times 5}{20} \text{ kJ mol}^{-1}$
 D $-\frac{60 \times 4.2 \times 5}{10} \text{ kJ mol}^{-1}$

- 11 Which equation correctly describes the reaction in which the ΔH is the standard enthalpy change of formation of water at 298 K?

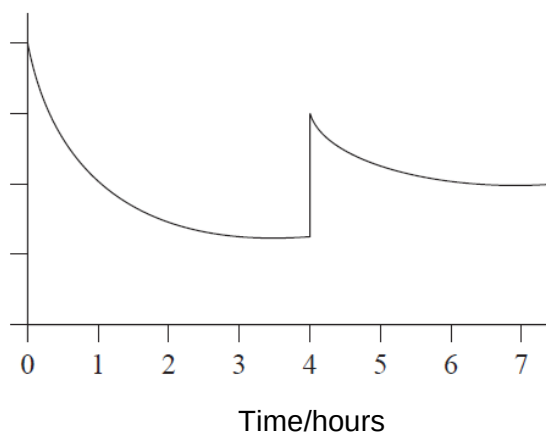


- 12 The reaction between phosphorus and hydrogen can result in the formation of phosphine as shown:



The graph shows the change in concentration of hydrogen for this reaction in which the system was disturbed after four hours.

Concentration of $\text{H}_2(\text{g})/\text{mol dm}^{-3}$



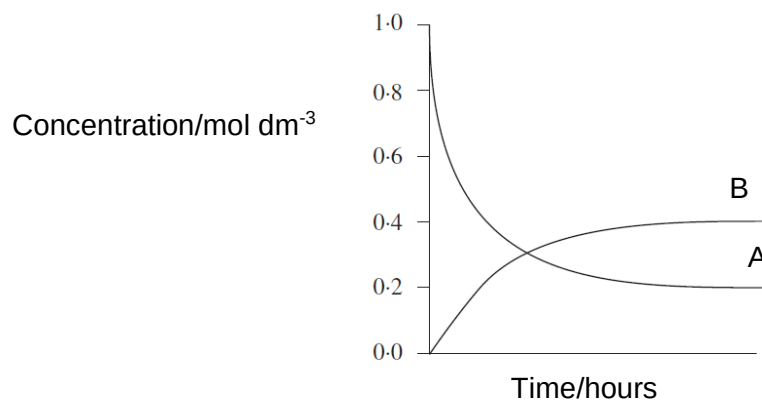
Which of the following could explain the change in the hydrogen concentration at time, $t = 4$ hours?

- A The volume of the reaction vessel was decreased.
 B A catalyst was added.
 C The pressure on the reaction mixture was decreased.
 D More phosphorus was added.

- 13 If compound **A** is heated, it decomposes according to the equation:



The following diagram shows the progress of the reaction.



What is the equilibrium constant for the reaction?

- A** 0.8 **B** 2.0 **C** 4.0 **D** 10.0
- 14 Technetium-99m is a radioactive isotope with a half-life of 6 hours. It is commonly used as a radiotracer in medical application.

Given that radioactive decay is a first order reaction, what percentage of the isotope would remain after 1 day?

- A** 6.25 **B** 12.5 **C** 25.0 **D** 93.8
- 15 The table below gives data for the reaction between **X** and **Y** at a constant temperature.

[X] /mol dm ⁻³	0.2	0.2	0.1	0.05
[Y] /mol dm ⁻³	0.2	0.1	0.1	D
relative rate	1.00	0.50	0.25	1.00

Which of the following statements is **not** correct?

- A** Keeping [X] constant, the rate of reaction is proportional to [Y].
B The reaction is first order with respect to [X].
C The overall order of reaction is 2.
D The value of **D** is 0.4.

- 16 25 cm³ of 0.1 mol dm⁻³ monobasic acid **X** is titrated with 50 cm³ of 0.1 mol dm⁻³ NaOH. The end point cannot be detected using screened methyl orange.

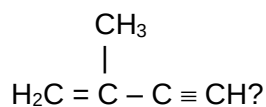
Which of the following is correct?

- A Acid **X** is a strong acid.
- B Initial pH of the solution is more than 1.00.
- C pH of the equivalence point is less than 7.
- D The solution acts as a buffer when 25.00 cm³ of NaOH has been added.
- 17 Which combination of solutions would **not** give a buffer solution?
- A 10 cm³ of 0.1 mol dm⁻³ CH₃CO₂H and 10 cm³ of 0.1 mol dm⁻³ CH₃CO₂Na.
- B 10 cm³ of 0.1 mol dm⁻³ NH₃ and 10 cm³ of 0.1 mol dm⁻³ NH₄Cl.
- C 10 cm³ of 0.1 mol dm⁻³ HCl and 10 cm³ of 0.1 mol dm⁻³ CH₃CO₂Na.
- D 10 cm³ of 0.1 mol dm⁻³ HCl and 20 cm³ of 0.1 mol dm⁻³ NH₃.
- 18 When an organic compound **X** was treated with phosphorus pentachloride, fumes of hydrogen chloride were evolved. When **X** was warmed with acidified aqueous potassium dichromate(VI), the solution turned green.

Which of the following was **X**?

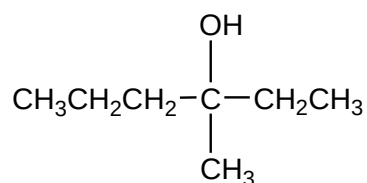
- A CH₃CH₂CHO
- B CH₃CH₂CO₂H
- C CH₃CH(OH)CH₃
- D CH₃CO₂CH₃

- 19 Which row in the table below has the correct number and type of bonds in



	Number of σ - bonds	Number of π - bonds
A	4	3
B	8	5
C	10	2
D	10	3

- 20 How many possible products (including geometric isomers) can be formed when the following compound is reacted with excess concentrated sulfuric acid at 180 °C?



- A** 3 **B** 4 **C** 5 **D** 6

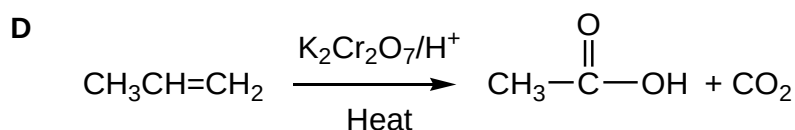
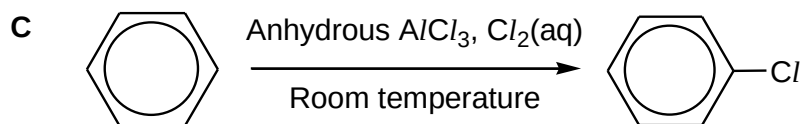
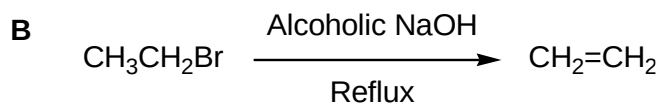
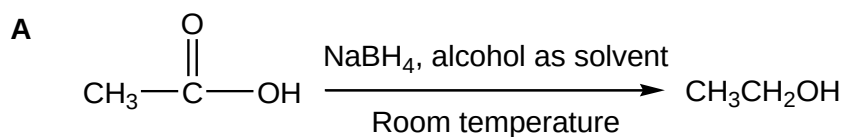
- 21 Alkane **P**, C_6H_{14} , was reacted with chlorine in the presence of ultraviolet light. Only three monochloro compounds of formula $\text{C}_6\text{H}_{13}\text{Cl}$ were obtained.

What is a possible structure of **P**?

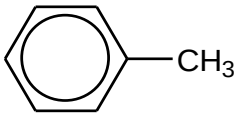
- A** $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$
B $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}(\text{CH}_3)_2$
C $\text{CH}_3\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_2\text{CH}_3$
D $\text{CH}_3\text{CH}_2\text{CH}(\text{CH}_3)\text{CH}_2\text{CH}_3$

22 The reaction conditions for four different transformations are given.

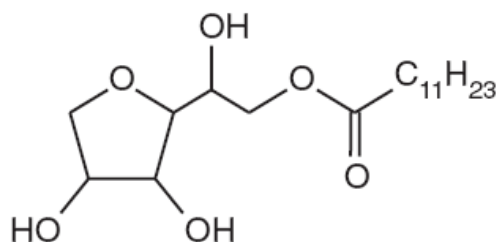
Which transformation has the correct conditions?



23 What will be observed when the following compounds are added to hot acidified potassium manganate(VII)?

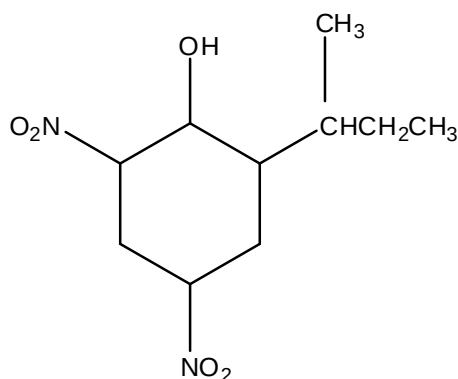
		$\text{CH}_3\text{CH}=\text{CH}_2$	$\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$
A	Solution turns colourless. No effervescence.	Solution turns colourless. Effervescence occurs.	Solution remains purple. No effervescence.
B	Solution remains purple. No effervescence.	Solution turns colourless. Effervescence occurs.	Solution turns colourless. No effervescence.
C	Solution turns colourless. Effervescence occurs.	Solution remains purple. No effervescence.	Solution remains purple. Effervescence occurs.
D	Solution remains purple. Effervescence occurs.	Solution remains purple. Effervescence occurs.	Solution turns colourless. No effervescence.

- 24 The demand for 'natural' shampoos and detergents has led to the development of more biodegradable detergents such as sorbitan monolaurate, which is made from plants. Its structure is shown below.



Which statement is correct?

- A There will be no colour change on heating the compound with acidified potassium dichromate(VI) solution.
 - B It gives a bright orange precipitate with 2,4-dinitrophenylhydrazine.
 - C It can react with concentrated sulfuric acid on heating.
 - D It liberates 3 moles of hydrogen gas with sodium metal.
- 25 Binapacryl is used as a fungicide. Its structure is shown below.



What deduction about binapacryl can be made from the above structure?

- A It reacts readily with iodine in aqueous sodium hydroxide.
- B It is acidic in nature.
- C It contains a tertiary alcohol.
- D It gives a ketone when oxidised.

Section B

For each of the questions in this section, one or more of the three numbered statements **1** to **3** may be correct.

Decide whether each of the statements is or is not correct (you may find it helpful to put a tick against the statements that you consider to be correct).

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

No other combination of statements is used as a correct response.

26 Which of the following molecules are polar?

1 CH_3OCH_3

2 CF_2Cl_2

3 C_2Cl_4

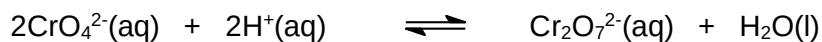
27 Which of the following properties **increase** from SiO_2 to P_4O_{10} to SO_3 ?

1 Melting point of the oxide

2 Covalent character

3 Oxidation number of the non-metal in the oxide

28 Chromate and dichromate ions form an equilibrium according to the following equation.



Which solution would increase the concentration of the chromate ion, CrO_4^{2-} , when added to the equilibrium mixture?

1 Sodium hydroxide

2 Hydrochloric acid

3 Sodium chloride

The responses **A** to **D** should be selected on the basis of

A	B	C	D
1, 2 and 3 are correct	1 and 2 only are correct	2 and 3 only are correct	1 only is correct

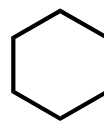
No other combination of statements is used as a correct response.

- 29** Chemists in the late 1800s knew that cyclic molecules existed, but the limitations on the ring sizes were unclear. Rings of all sizes from three through thirty and beyond can now be prepared easily.

Cyclopropane is known to be less stable than cyclohexane.



Cyclopropane

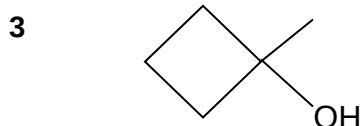


Cyclohexane

Which reasons explain the difference in stability?

- 1** Delocalisation of π electrons strengthens the carbon–carbon bonds in cyclohexane.
 - 2** The C-C-C bond angle in cyclopropane experiences a large deviation from the ideal value of 109.5° .
 - 3** The carbon atoms of cyclohexane are not in the same plane, allowing bond angles to be near the ideal value.
- 30** The alcohol $C_5H_{10}O$, may be oxidised by acidified aqueous potassium dichromate(VI) to a compound C_5H_8O . Which of the following could be the structural formula of the alcohol?

- 1** $CH_3CH = CHCH_2CH_2OH$
- 2** $CH_2 = C(CH_3)CH(OH)CH_3$



Answers to 2012 TJC H1 Chem Prelim Paper 1

1	2	3	4	5	6	7	8	9	10
D	B	C	B	C	C	A	B	C	C
11	12	13	14	15	16	17	18	19	20
B	A	C	A	D	B	C	C	D	C
21	22	23	24	25	26	27	28	29	30
A	B	A	C	D	B	C	D	C	B